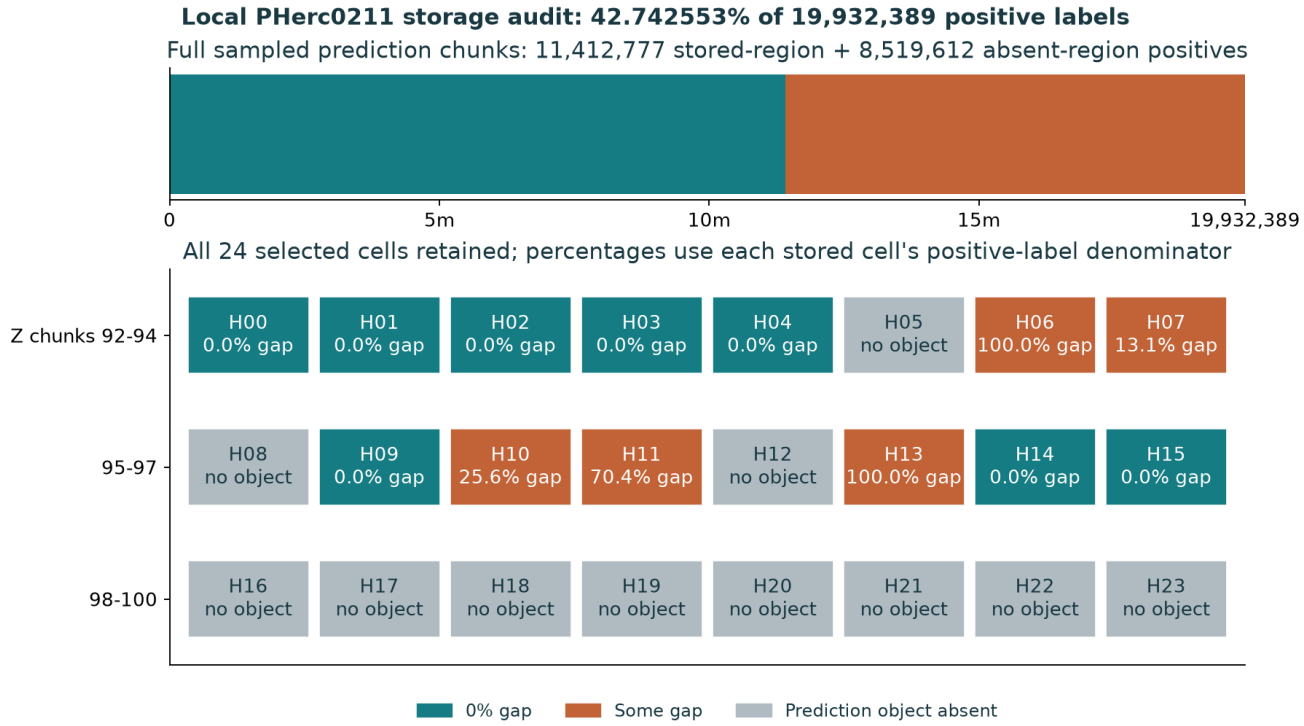
What the archived evidence supports

A reproducible local audit of prediction labels and masked-CT storage

Computational surface predictions can be read more confidently than their source-data support justifies. ARGUS makes a limited part of that relationship inspectable: positive prediction labels versus masked-CT chunk storage on nominal matching indices.

Selected local cells	Positive prediction labels	In absent masked-CT storage
24 (13 stored, 11 absent)	19,932,389	8,519,612 / 42.742553%



Method: decode captured original label chunks; derive each CT-chunk intersection; reconcile storage states with archived HTTP receipts; retain all sampled cells and the denominator. The six earlier selected examples are reported separately.

Claim limit: these are label/storage observations. Missing masked storage does not establish missing original CT, invalid papyrus, physical registration, ink or writing. The percentage is local and positive-voxel weighted, not whole-scroll prevalence.

Why I began ARGUS

I'm Spencer Tropea, an independent creator and researcher from Ontario, Canada. My background is in communications, and my interests in history and literature led me to the possibility of recovering writing thought lost for centuries.

ARGUS began with that possibility and a practical question: how do we know that what our software shows is supported by the evidence? I have developed it iteratively with AI-assisted software development, open-source research, experimental tests and separate checks of captured data. I am contributing an audit workflow, learning from techniques and expertise established by others.

The name recalls Argus Panoptes, the many-eyed watcher of Greek mythology: a reminder to inspect carefully and keep uncertainty visible.

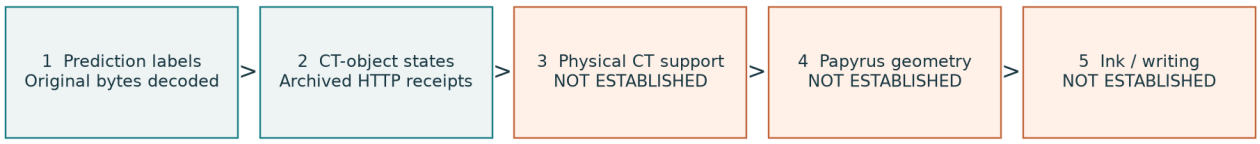
These scrolls were once people's words and possessions. I found myself wanting, symbolically, to ask their permission to look; for me, that means approaching their surviving intellectual lives with care, without a sense of entitlement.

I welcome scientific criticism, independent reproduction and collaboration. My aim is to help preserve knowledge through useful, accountable work, wherever that contribution belongs in the existing research community.

A practical contribution, open to correction

The work was developed with AI-assisted software tools and existing open-source research. Separate counting implementations agree on these archived inputs; no external researcher reproduction, institutional endorsement or server attestation is claimed.

Evidence levels: only the first two are established here



Nominal index correspondence is not independently verified physical registration.

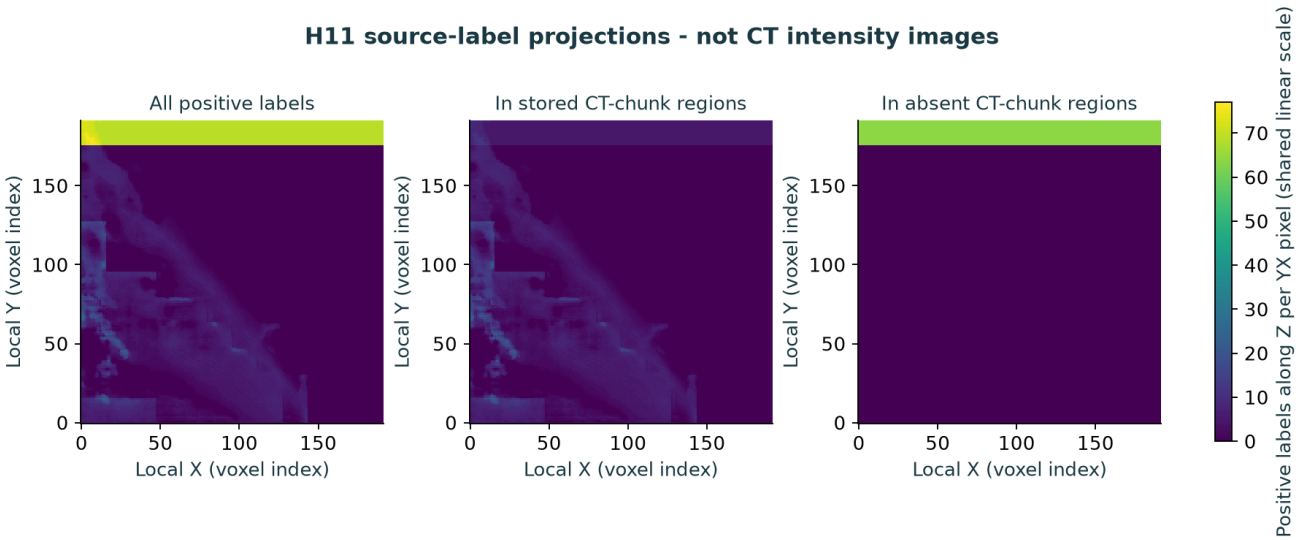
Established here: original prediction-label counts and archived masked-CT storage states. Physical CT support, verified sheet geometry and ink identification remain separate evidence gates.

The 24-cell local survey

The deterministic plan regenerates from 423 eligible prediction cells intersecting the historically selected R04 halo, excluding 18 prior survey keys. Eight cells are hash-ranked within each of three fixed Z strata. The archived hash verifies content; it does not independently prove when the plan was fixed or rule out historical outcome-peeking.

Z chunk stratum	Stored / absent predictions	Positive labels	In absent CT storage
92-94	7 / 1	9,332,548	2,789,605
95-97	6 / 2	10,599,841	5,730,007
98-100	0 / 8	0	0

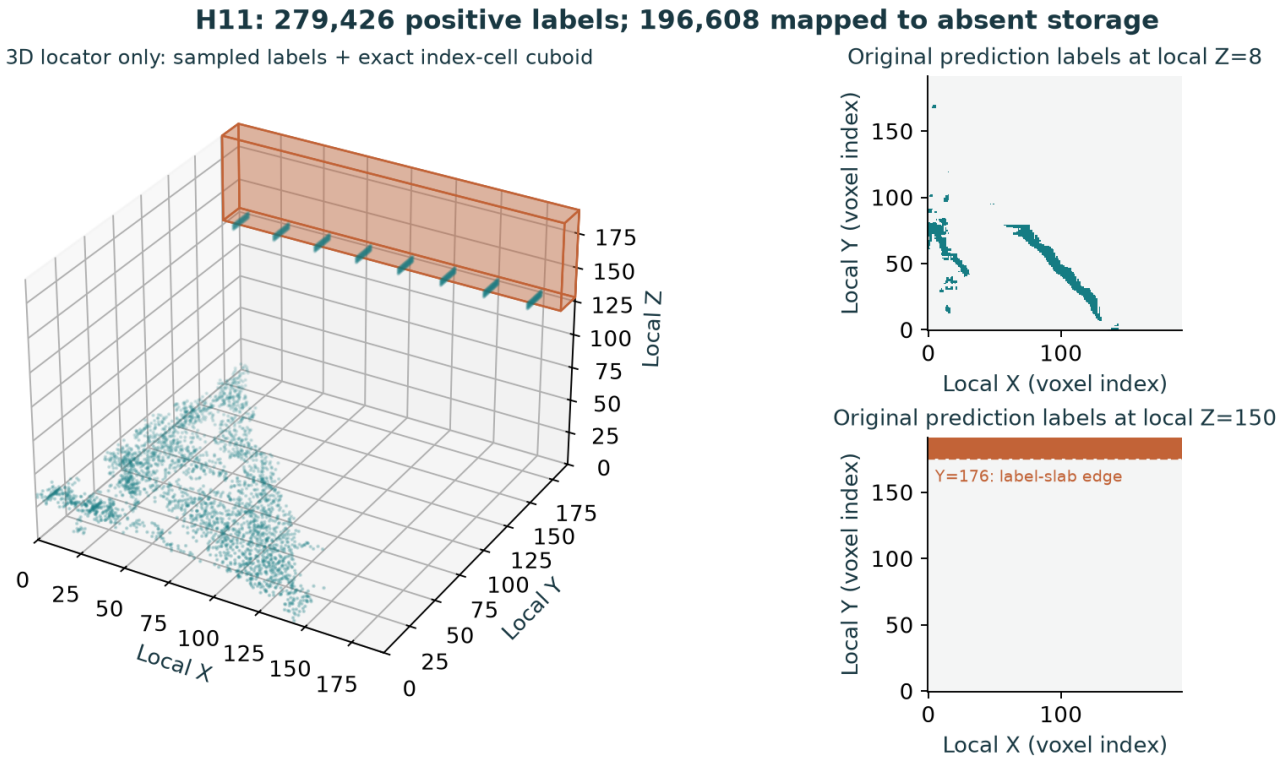
All 24 selected locations and 186 distinct CT keys were accounted for. The archive contains 212 request receipts: 26 prediction/metadata GETs and 186 CT HEADs. CT receipts record 75 stored and 111 absent objects.



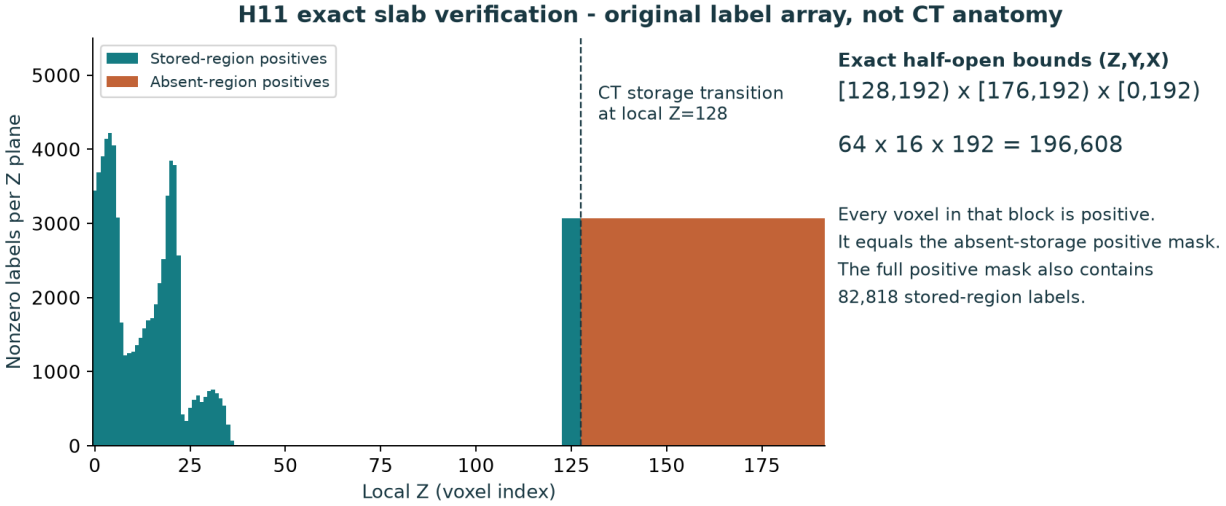
H11 projections illustrate source-label content, not CT anatomy. Color intensity is a count of positive labels along Z per YX pixel. Source: unchanged original H11 Blosc bytes, SHA256 recorded with the figure.

Counts use FULL 192-cubed prediction chunks selected for intersecting the halo. They are not halo-clipped or R12-specific. Spatial dependence and the historical frame prevent a whole-scroll or classifier-performance inference.

H11: the block is in the label array

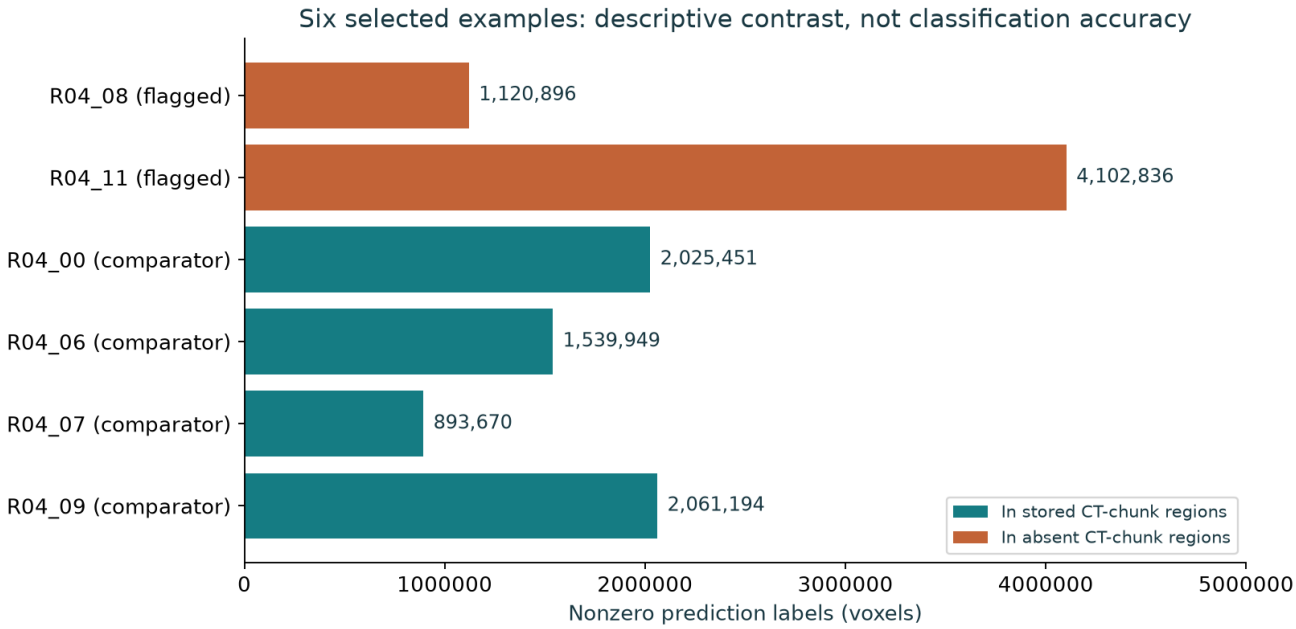


Original prediction chunk 96/32/23 contains 279,426 positive labels: 82,818 in stored CT-chunk regions and 196,608 in absent regions. Exactly the absent-region positive mask equals local Z[128,192), Y[176,192), X[0,192). The full positive mask also includes the 82,818 other labels.



Correction: Y=176 is the label slab edge. The archived CT-object transition is at local Z=128. Coarse 3D models can expand edges through max pooling; the original voxel array, not the model, establishes the exact count and bounds. Cause and physical significance remain unresolved.

Descriptive cases and prior art



The six earlier cases were selected after preliminary observations. Two flagged chunks contribute 5,223,732 absent-region positives; four contrasting comparators contribute 6,520,264 stored-region positives. They demonstrate counting behavior and must not be pooled with the new survey to estimate rates.

Complementary value must be demonstrated

Schurkai's vesuvius-phantom-audit already offers scroll-wide chunk audits, voxel-level checks and cleanup. A matched comparison of its pinned pure overlap function with ARGUS agrees at all 24 sampled cells. This establishes overlap in functionality; it is not a full runtime or end-to-end benchmark.

Existing capability	ARGUS candidate addition / limit
Chunk and voxel checks; cleanup	Compact fixed-sample acquisition, local receipts and replayable partitions
Native CT intensity checks in upstream cleanup	Current submission uses object states; no independently registered CT-intensity validation
Published upstream code	Reviewed source release; no demonstrated adoption or practical superiority

Pinned prior-art commit: ab9ace908a1644c32ce0206e3eecfefba2c4bc81. Function agreement is recorded in docs/UPSTREAM_SOURCE_RECORD.json and the prior local review. No new defect category or superiority is claimed.

Technical review: fixes and limits

Review finding	Revised behavior
Oversize/budget errors could return stale HTTP 200	Failed bodies remain unknown; regression tests pass
Capture/error output could overwrite evidence	Exclusive creation and separate diagnostic ZIP
Unknown state and duplicate partition risks	Unknown normalization; disjointness and partial-state checks
Conflicting versions and missing commands	Reviewed source 0.4.3; three documented capture replays
Misleading figure boundary and mask property	Correct Z=128 storage transition; precise slab-mask equality

The received nine tests pass; the reviewed source passes 16 tests on Windows Python 3.12.14 with NumPy and Blosc2. The supplied row-wise verifier and a separately implemented review audit reproduce the central totals. Tests do not certify every inherited library path or substitute for clean-platform checks.

Reproducibility has a defined scope

Source compressed bytes, metadata fingerprints, local receipt records, deterministic selection and per-intersection counts are available for replay. SHA256 establishes internal consistency, not a third-party signature or a proof of capture chronology. Physical registration remains unverified.

Code and data have different rights

Original ARGUS software carries an MIT grant. The AWS Vesuvius CT registry identifies CC BY-NC 4.0 data; exact prediction-asset notices must be checked before public redistribution. Original source data, derived figures and models are not relicensed by the software grant.

Clean Windows/Linux Python 3.11 and 3.13 checks are configured in CI; see docs/VALIDATION.md and the exact commit checks for their actual status. A full upstream benchmark, demonstrated user benefit, physical registration and external researcher reproduction remain open.

November-December: conditional progress

The aim is one useful, maintained capability in the existing research ecosystem. These are proposed milestones, not promises of discovery or completed community integration.

Window	Deliverable / decision gate
Nov 2-6	Matched end-to-end benchmark; distinguish useful addition from duplication
Nov 9-13	Independent transforms and mask-semantics review: verified / failed / unresolved
Nov 16-20	One traceable downstream inspection or processing decision
Nov 23-30	Frozen independent region/scroll; preserve all failures and denominators
Dec 1-11	Clean platforms, schema and resource checks; invite volunteer replay
Dec 14-18	Technical note and maintain / upstream / retire decision
Dec 21-31	Buffer for documentation, feedback and maintenance

Opportunities for Collaboration

Review the prediction-to-CT assumptions; reproduce H11 or one fixed case; test another dataset; recommend an existing workflow integration point; advise whether receipt or partition exports belong upstream.

> If registration fails, preserve only storage-state claims. If an existing tool supplies equivalent value, prefer an upstream contribution or retire duplication. R12 stays outside this critical path and Gate 7F remains blocked.

I welcome scientific criticism, independent reproduction and collaboration. No endorsement, private data access, computing allocation or appointment is requested.

Reproduce the public source release

From the repository root, install the verification extra and replay the retained historical capture. Offline replay makes no network requests. The README documents the historical capture rights limits and all three counting commands.

```
python -m pip install ".[verification]"
```

CAPTURE = ARGUS_04_HELDOUT_CAPTURE.zip

```
python -O tools/review_release.py CAPTURE --out RELEASE_REPLAY
```

```
python -m unittest discover -s tests -v
```

Source release and research limits

Reviewed source is prepared for the existing Troptrops/ARGUS repository. Source, protocol, reports and figures can be inspected individually. The README and CI report exact release status; a full benchmark and one downstream user decision would strengthen the prize case. Asset-specific prediction rights remain unresolved; no new raw data is introduced.

The author authorized source publication. Competition submission and outreach are separate decisions and have not been performed here. Source publication does not establish scientific acceptance or an award. Historical capture preservation is not a blanket grant to redistribute prediction bytes.

Sources and attribution

Dataset: Vesuvius Challenge - CT Scans of Herculaneum Papyri. Newer PHerc0211 scan and m7 prediction data, credited to the Vesuvius Challenge creators and maintainers. scrollprize.org/data

Dataset terms: AWS Vesuvius dataset registry (CC BY-NC 4.0; check asset-specific terms).

Prior work: github.com/Schurkai/vesuvius-phantom-audit; ScrollPrize/villa and their contributors.

Prize rules and actual form entry point: scrollprize.org/prizes. Deadline checked 28 October: 31 October 2026, 11:59 pm Pacific.

> Look carefully. Verify what you can. Be honest about what you cannot yet see.

Public repository: https://github.com/Troptrops/ARGUS. Evidence: evidence/; protocol: protocol/; source: argus_support/ and tools/; verification: docs/VALIDATION.md; hashes: FILE_SHA256.txt. Historical reports are marked separately under research/history/.